

BUILT_IN FUNCTIONS

- 1) Write a query using the Production.Product table displaying the ProductID, Color, and Name columns. If the Color column contains a NULL value, replace the color with 'No Color'.

USE AdventureWorks_Practice;

GO

SELECT

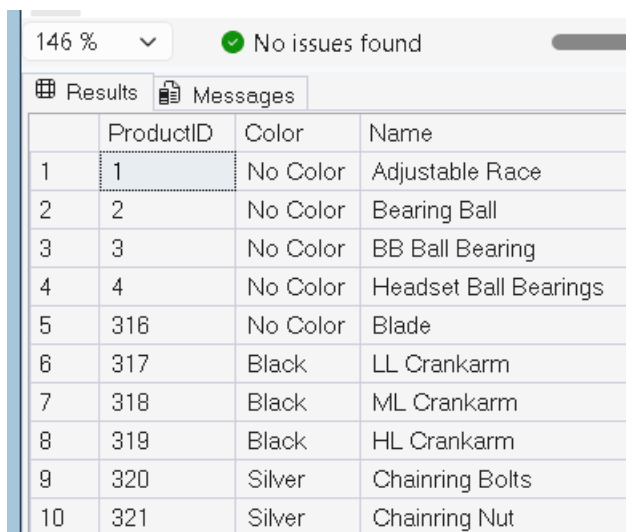
ProductID,

ISNULL(Color, 'No Color') AS Color,

Name

FROM Production.Product;

Output



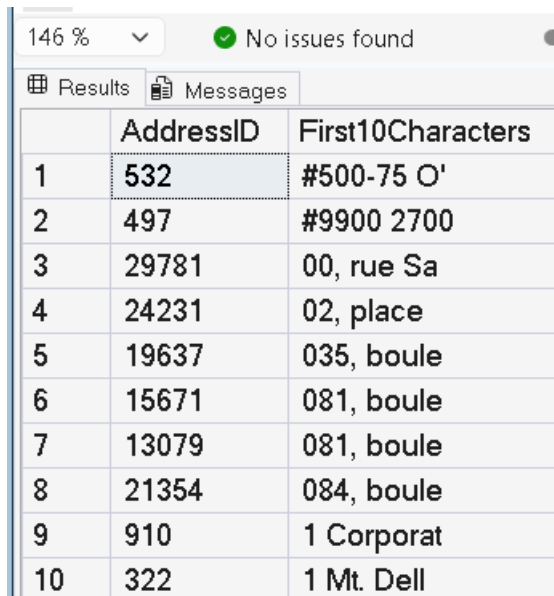
146 % No issues found

	ProductID	Color	Name
1	1	No Color	Adjustable Race
2	2	No Color	Bearing Ball
3	3	No Color	BB Ball Bearing
4	4	No Color	Headset Ball Bearings
5	316	No Color	Blade
6	317	Black	LL Crankarm
7	318	Black	ML Crankarm
8	319	Black	HL Crankarm
9	320	Silver	Chainring Bolts
10	321	Silver	Chainring Nut

2) Write a query that displays the first 10 characters of the AddressLine1 column in the Person.Address table.

```
SELECT  
  
    AddressID,  
  
    LEFT(AddressLine1,10) AS First10Characters  
  
FROM Person.Address;
```

Output



	AddressID	First10Characters
1	532	#500-75 O'
2	497	#9900 2700
3	29781	00, rue Sa
4	24231	02, place
5	19637	035, boule
6	15671	081, boule
7	13079	081, boule
8	21354	084, boule
9	910	1 Corporat
10	322	1 Mt. Dell

3) Write a query using the Sales.SpecialOffer table that multiplies the MaxQty column by the DiscountPct column. If MaxQty is NULL, replace it with the value 10. Include the SpecialOfferID and Description columns.

```
SELECT  
  
    SpecialOfferID,
```

```

Description,
ISNULL(MaxQty,10) AS MaxQty,
DiscountPct,
ISNULL(MaxQty,10) * DiscountPct AS Result
FROM Sales.SpecialOffer;

```

Output

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 Results

 Messages

	SpecialOffer...	Description	MaxQty	Discount...	Result
1	1	No Discount	10	0.00	0.00
2	2	Volume Discount 11 to 14	14	0.02	0.28
3	3	Volume Discount 15 to 24	24	0.05	1.20
4	4	Volume Discount 25 to 40	40	0.10	4.00
5	5	Volume Discount 41 to 60	60	0.15	9.00
6	6	Volume Discount over 60	10	0.20	2.00
7	7	Mountain-100 Clearance S...	10	0.35	3.50
8	8	Sport Helmet Discount-2002	10	0.10	1.00
9	9	Road-650 Overstock	10	0.30	3.00
10	10	Mountain Tire Sale	10	0.50	5.00

4) Write a query that calculates the number of days between the OrderDate and ShipDate using the Sales.SalesOrderHeader table. Include the SalesOrderID, OrderDate, and ShipDate columns.

```

SELECT
    SalesOrderID,
    OrderDate,

```

```

ShipDate,

DATEDIFF(DAY, OrderDate, ShipDate) AS DaysTaken

FROM Sales.SalesOrderHeader;

```

Output

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Results Messages					
	SalesOrderID	OrderDate	ShipDate	DaysTaken	
1	43659	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
2	43660	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
3	43661	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
4	43662	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
5	43663	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
6	43664	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
7	43665	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
8	43666	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
9	43667	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	
10	43668	2011-05-31 00:00:00.000	2011-06-07 00:00:00.000	7	

5) Write a query that displays the year and numeric month of each OrderDate. Include the SalesOrderID and OrderDate columns.

```

SELECT

SalesOrderID,

OrderDate,

YEAR(OrderDate) AS OrderYear,

MONTH(OrderDate) AS OrderMonth

FROM Sales.SalesOrderHeader;

```

Output

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 Results

 Messages

	SalesOrderID	OrderDate	OrderYear	OrderMonth
1	43659	2011-05-31 00:00:00.000	2011	5
2	43660	2011-05-31 00:00:00.000	2011	5
3	43661	2011-05-31 00:00:00.000	2011	5
4	43662	2011-05-31 00:00:00.000	2011	5
5	43663	2011-05-31 00:00:00.000	2011	5
6	43664	2011-05-31 00:00:00.000	2011	5
7	43665	2011-05-31 00:00:00.000	2011	5
8	43666	2011-05-31 00:00:00.000	2011	5
9	43667	2011-05-31 00:00:00.000	2011	5
10	43668	2011-05-31 00:00:00.000	2011	5

6) Write a query using the Sales.SalesOrderHeader table that displays the SubTotal rounded to 2 decimal places. Include the SalesOrderID.

SELECT

SalesOrderID,

ROUND(SubTotal,2) AS RoundedSubTotal

FROM Sales.SalesOrderHeader;

Output

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Results Messages

	SalesOrderID	RoundedSubTotal
1	43659	20565.62
2	43660	1294.25
3	43661	32726.48
4	43662	28832.53
5	43663	419.46
6	43664	24432.61
7	43665	14352.77
8	43666	5056.49
9	43667	6107.08
10	43668	35944.16

7) Write a statement that generates a random number between 1 and 10 each time it is run.

SELECT FLOOR(RAND() * 10) + 1 AS RandomNumber;

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Results Messages

	RandomNumber
1	6